TECHNICAL REPORT

A Visual Study of E-Commerce Orders, Delivery & Customer Satisfaction

Growing a marketplace without breaking the customer experience

Data Visualization Design Project
Indian Institute of Technology Madras - BS Degree

Group 10

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Repository: github.com/surya101p/DVD_23f2004625

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1. Executive Summary

This report studies the full journey of an order on a large Brazilian online marketplace from placement, payment and seller handling, through freight and delivery, to the review the customer finally leaves. Across **96,469 delivered orders** worth **BRL 15.5M** placed by **93,349 unique customers**, the average review score is **4.16/5** and only **8.1%** of orders arrive late. The marketplace is healthy on average but the averages hide the single most important lever in the business.

Headline Finding

> **Delivery timing is the dominant driver of satisfaction.** On-time orders average 4.29 ⋆ with a 9% negative-review rate. Late orders average 2.57 ⋆ with a 54% negative-review rate and 46% one-star rate. A late delivery does not shave a fraction off satisfaction, it roughly **halves** the rating and multiplies bad reviews five-to-six fold.

What to Do About It

- **Protect the delivery promise before expanding it.** On-time delivery, not price or product, is the metric leadership should manage most tightly as the marketplace grows.
- **Intervene on 11 high-revenue, low-satisfaction categories** (watches & gifts, bed/bath/table, computers & accessories, furniture décor and others) that earn significantly but rate below the marketplace median.
- **Target the regional delivery gap.** Late-delivery pain is concentrated in the North/Northeast (Alagoas 23.9%, Maranhão 19.6%) where ratings are lowest, while São Paulo, 42% of all orders, is fast and well-rated.
- **Seller risk is concentrated, not systemic.** Of 429 high-volume sellers, only one is genuinely high-risk, so quality can be improved through targeted support rather than a broad crackdown.
- **Spend acquisition budget on efficient channels:** paid and organic search convert leads to sellers at ∼12%, while social converts at only 5.6% despite comparable lead volume.

2. Business Problem & Analytical Approach

2.1. Context

The client is an online marketplace that connects thousands of small and medium sellers to customers across Brazil, a large and geographically spread-out country, and provides the logistics network that fulfils their orders. Leadership has one goal: **grow the marketplace without breaking the customer experience**. Every growth decision trades off against satisfaction. Onboard sellers aggressively and gross sales rise, but late deliveries, poorly rated products and bad reviews start driving customers away. Crack down on every slow seller and quality improves, but the catalogue shrinks and growth stalls.

Today these calls are made on intuition.

2.2. Business Goal, Restated

Turn the end-to-end order journey into a clear visual account of what makes a customer happy versus what earns a one-star review, so that leadership can see, by category, seller, region and delivery pattern, exactly where the marketplace should invest to grow and where it must intervene to protect the customer experience.

2.3. Six Analysis Questions

We translated the business goal into six concrete, answerable questions. Each is addressed with an explanatory visual in Section 6.

#	Question	Answered in
Q1	Does delivery performance drive customer satisfaction, and by how much?	Section 6.1
Q2	Which product categories should the company invest in vs. intervene on?	Section 6.2
Q3	Where, geographically, is the marketplace most at risk?	Section 6.3
Q4	Is poor satisfaction a systemic seller problem or concentrated in a few?	Section 6.4
Q5	Do shipping cost and freight relate to how customers rate an order?	Section 6.5
Q6	Which marketing channels most efficiently acquire new sellers?	Section 6.6

Table 1. The six analysis questions derived from the business goal.

2.4. Approach & Methodology

The project followed a five-stage workflow: (1) understand the business goal and data; (2) clean and prepare the data and engineer analytical features; (3) build exploratory visuals to uncover structure; (4) build explanatory visuals that answer the questions above; and (5) present the story through this technical report, an interactive dashboard and a final presentation.

- **Tools:** Python 3 with `pandas` and `NumPy` for data preparation; `matplotlib` and `seaborn` for report figures; Streamlit with Plotly for the interactive dashboard. Python was chosen for its mature data-handling ecosystem and reproducibility, and Streamlit for turning the same analytical tables directly into an interactive web app with minimal overhead.
- **Analytical grain:** the unit of analysis is the *delivered order*, only these have complete timestamps and meaningful delivery outcomes, then aggregated to category, seller, and state level.
- **Reproducibility:** the pipeline is fully scripted end-to-end (see the Technical Appendix). One script cleans raw CSVs and writes analytical tables; a second regenerates all figures. The dashboard reads the same tables.

- The findings are also available as an **interactive Streamlit dashboard** at the repository link, structured across five tabs: KPI Header, Category Strategy, Delivery & Satisfaction, Regional, and Seller Performance.

3. Dataset Overview

3.1. Sources and Scale

The project uses the **Olist Brazilian E-Commerce** dataset paired with a companion **Marketing Funnel** dataset, 11 CSV files across two folders. The e-commerce folder (9 files) covers the full transactional lifecycle of orders on the marketplace; the marketing folder (2 files) covers seller-acquisition leads and closed deals.

- **~99,441 orders** placed between September 2016 and October 2018 (the bulk of volume in 2017–2018), by **~96,096 unique customers**.
- **~112,650 line items** across those orders, sold by **~3,095 sellers**, covering **~32,951 products** in **71 (+2 unmapped)** categories.
- **~103,886 payment records** and **~99,224 order reviews** on a 1–5 star scale.
- **~1.0M raw geolocation records** mapping ZIP-code prefixes to latitude/longitude (heavily duplicated).
- **Marketing funnel:** 8,000 Marketing Qualified Leads (MQLs), of which 842 became closed deals (sellers who joined), spanning June 2017–November 2018.

3.2. Data Structure

The tables link through common ID keys into a relational structure centred on the orders table.

File	Grain	Key columns	Links to
`customers`	1 row / customer	`customer_id`, `customer_unique_id`, zip / city / state	orders
`orders`	1 row / order	`order_id`, `order_status`, 5 timestamps	customers, items, payments, reviews
`order_items`	1 row / item in order	`order_id`, `product_id`, `seller_id`, price, freight	orders, products, sellers
`order_payments`	1+ rows / order	`order_id`, `payment_type`, installments, value	orders
`order_reviews`	1 row / review	`review_id`, `order_id`, `review_score`, text	orders
`products`	1 row / product	`product_id`, category, dims, weight, photos	order_items, translation

File	Grain	Key columns	Links to
`category_-translation`	1 row / category	category name → English	products
`sellers`	1 row / seller	`seller_id`, zip / city / state	order_items, closed_deals
`geolocation`	1 row / zip lookup	zip prefix, lat, lng, city, state	customers / sellers via zip
`marketing_-qualified_-leads`	1 row / lead	`mql_id`, first-contact date, origin	closed_deals
`closed_deals`	1 row / won deal	`mql_id`, `seller_id`, won date, segment	mql, sellers

Table 3. The eleven source tables, their grain and how they join.

3.3. Key Variable Groups

- **Time:** the five order timestamps (purchase → approval → carrier hand-off → customer delivery → estimated delivery) enable delivery-performance and SLA analysis.
- **Money:** price, `freight_value`, `payment_value`, installments and payment type support revenue, AOV and payment-behaviour analysis.
- **Satisfaction:** `review_score` (1–5) plus free-text comments.
- **Geography:** customer and seller state/city, joinable to geolocation for mapping.
- **Product:** category (English via the translation table), weight, dimensions and photo count.
- **Funnel/acquisition:** lead origin, landing page, declared business size and won date support seller-acquisition analysis.

4. Data Preparation & Cleaning

Before analysis, the raw tables were cleaned, type-corrected, checked for quality issues, and combined into a single order-level analytical dataset. This section documents each step; the corresponding code is in the Technical Appendix.

4.1. Type Conversion & De-duplication

- All ten date/time columns across the orders, reviews and marketing tables were parsed to proper datetime types (with invalid values coerced to null) so that durations could be computed.
- The geolocation table contained **261,831 fully duplicated rows**; duplicates were removed and coordinates treated as one representative point per ZIP prefix.

4.2. Missing Values

Aside from optional or structurally expected gaps, the standout issue is the seller-intake fields in `closed_deals`, which are 90%+ empty and must be treated as sparse optional attributes rather than core features.

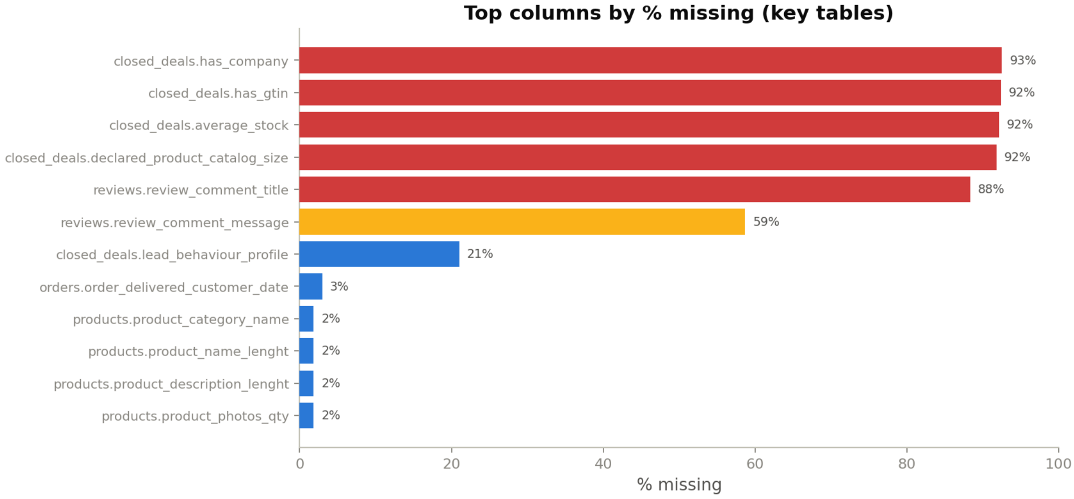

Figure 1. Top columns by percentage missing across the key tables. Free-text review fields and closed-deal intake fields dominate.

Table	Field(s)	Missing	Treatment
orders	approved / carrier / delivered dates	160 / 1,783 / 2,965	Expected, non-delivered orders; excluded from delivery timing
reviews	comment title / message	87,656 / 58,247	Expected, optional free text; filled with empty string
products	category + 3 attributes	610 each	Labelled 'unknown'; dimensions imputed with median
mql	origin	60	Filled with 'unknown'
closed_deals	company / gtin / stock / catalog size	773–779 of 842 (∼92%)	Treated as sparse; not used as core features

Table 5. Missing-value summary and how each case was handled.

4.3. Outliers

Item price is strongly right-skewed (mean BRL 120.65 vs. median BRL 74.99, max BRL 6,735); about 7.5% of rows sit above the IQR upper fence. This is consistent with a marketplace selling both everyday and premium goods rather than a handful of data errors, so prices were retained but examined on a log scale. Payment value shows a similar skew, with a handful of logically odd zero-value payments. Four products carry

0 g weight and a small number of orders take 60–90+ days to deliver, flagged and excluded from timing analysis where they distort durations.

4.4. Logical & Referential Inconsistencies

- **1,359 orders** show the carrier hand-off date earlier than the approval date, and **23 orders** show customer delivery earlier than carrier hand-off, sequence violations excluded from timing analyses.
- **7,827 of 96,476 delivered orders (∼8%)** arrived after the estimated delivery date, not an error but the natural late-delivery / SLA-miss signal that Section 6 builds on.
- **2 product categories** ('portateis_cozinha_e_preparadores_de_alimentos' and 'pc_gamer') have no English translation and were given a fallback label.

4.5. Feature Engineering

Cleaned tables were merged into an order-level analytical dataset, adding the derived fields that make the analysis possible:

Derived field	Definition
`seller_handling_days`	Days from purchase to carrier hand-off (seller-controlled leg)
`shipping_days`	Days from carrier hand-off to customer delivery (logistics-controlled leg)
`delivery_days`	Total days from purchase to customer delivery
`delivery_delay_days`	Actual minus estimated delivery (negative = arrived early)
`is_late`	1 if delivered after the marketplace's estimated delivery date
`delivery_performance`	Categorical: On Time / 1–3 days late / 4–7 days late / >7 days late
`is_negative_review`	1 if review score ≤ 2
`is_one_star`	1 if review score = 1
`order_value`	Product value + freight for the order
`strategy`	Category quadrant assignment: Invest / Intervene / Maintain / Review

Table 6. Engineered fields used throughout the analysis.

5. Exploratory Data Analysis

Exploratory analysis establishes the shape and structure of the marketplace before we test specific questions. Each subsection pairs a visual with the structure it reveals.

5.1. Order Volume Over Time

Monthly purchase volume traces the marketplace's growth: negligible activity through most of 2016, a steep ramp from early 2017, and a plateau around 6,000–7,000 orders per month through 2017–2018, with a clear spike around Black Friday (November 2017).

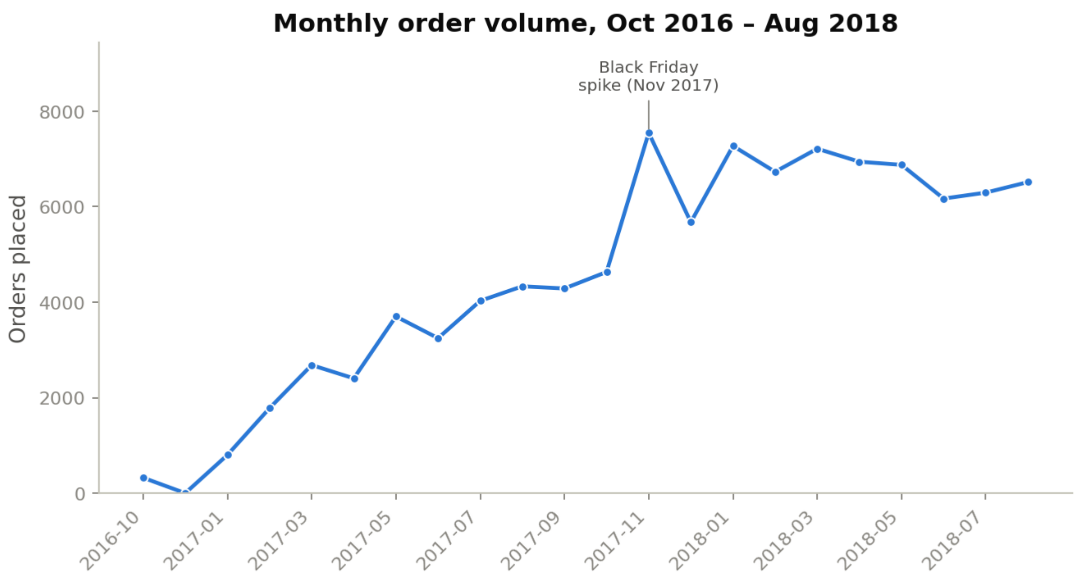

Figure 2. Monthly order volume, Oct 2016 – Aug 2018 (sparse trailing months excluded).

5.2. Order Status

The order lifecycle is overwhelmingly successful: delivered orders dwarf every other status combined, so a log scale is used to keep the small failure states visible.

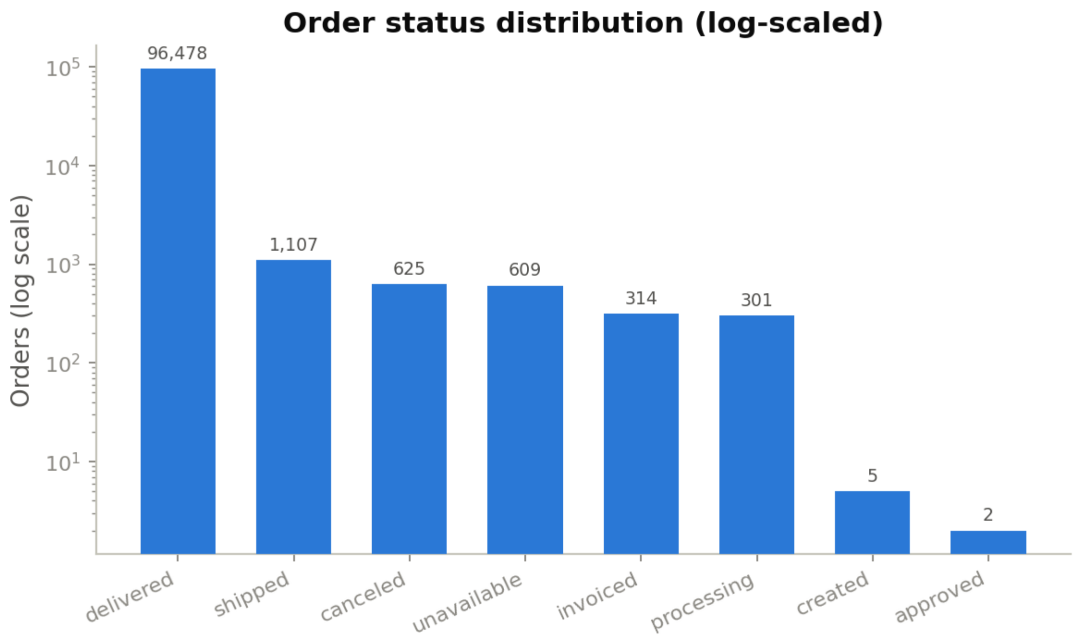

Figure 3. Order status distribution (log-scaled). ‘Delivered’ dominates the lifecycle.

5.3. Customer Satisfaction

Review scores are bimodal and skew positive: 58% are 5-star, but 1-star reviews (12%) form the second-largest group, the classic pattern of dissatisfied customers being the most motivated to rate. This is the outcome variable the rest of the report explains.

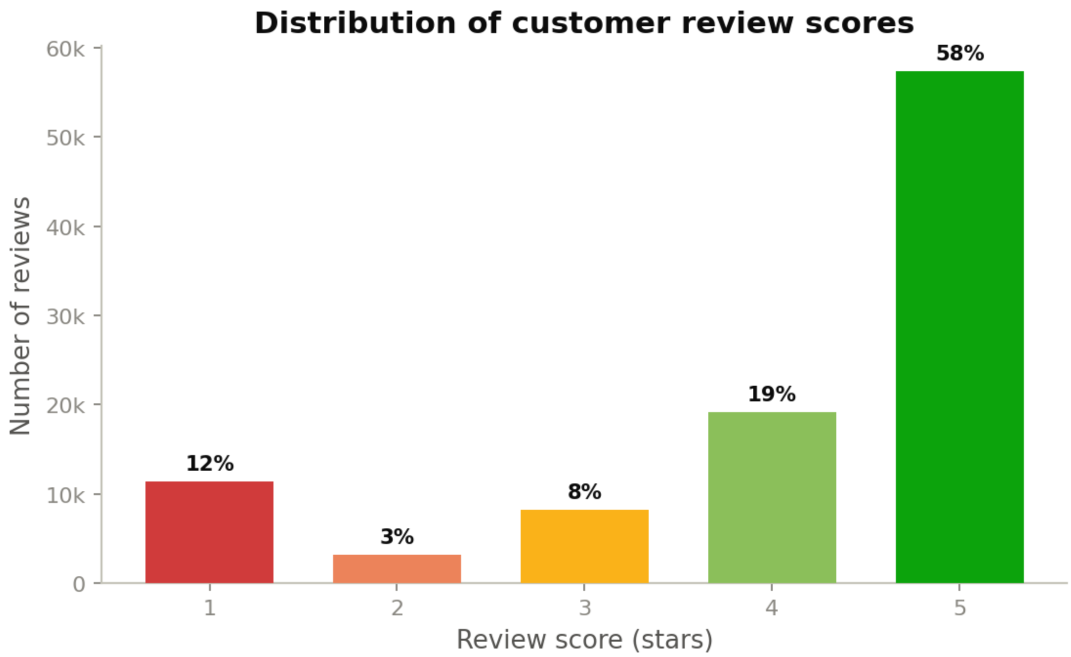

Figure 4. Distribution of review scores with percentage share. Positive skew, but a heavy one-star tail.

5.4. Payments

Credit card is the dominant payment method (74% of records), followed by boleto, Brazil's bank-slip method, at 19%, then vouchers and debit cards.

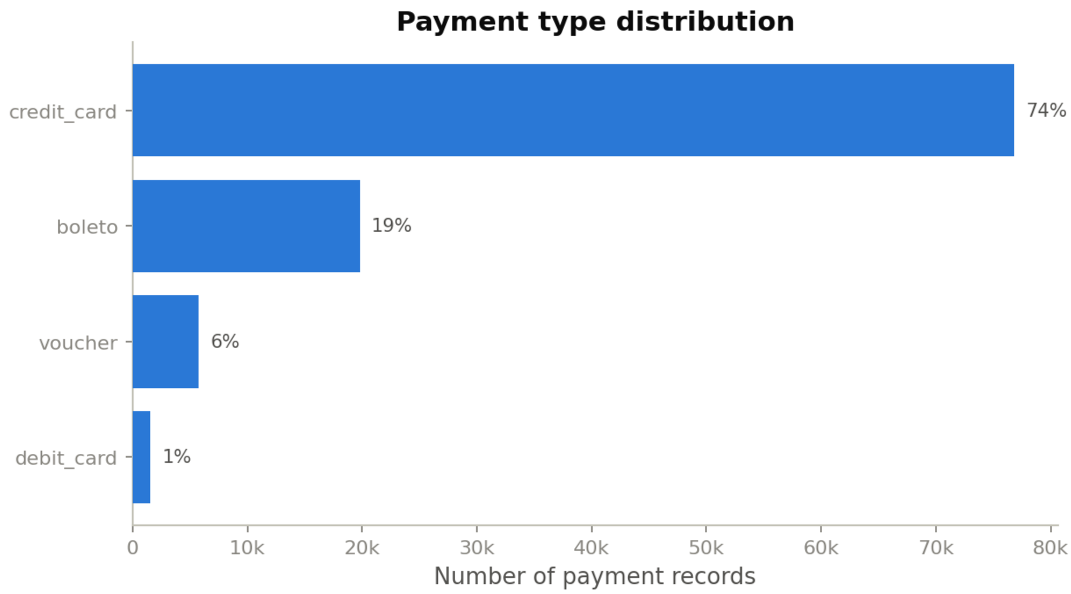

Figure 5. Payment type distribution. Credit card and boleto account for over nine in ten payments.

5.5. Price Distribution & Outliers

Item price is approximately log-normal with a long right tail: the log-scaled histogram is roughly symmetric, and the boxplot shows a continuous tail of premium items rather than a few extreme flukes.

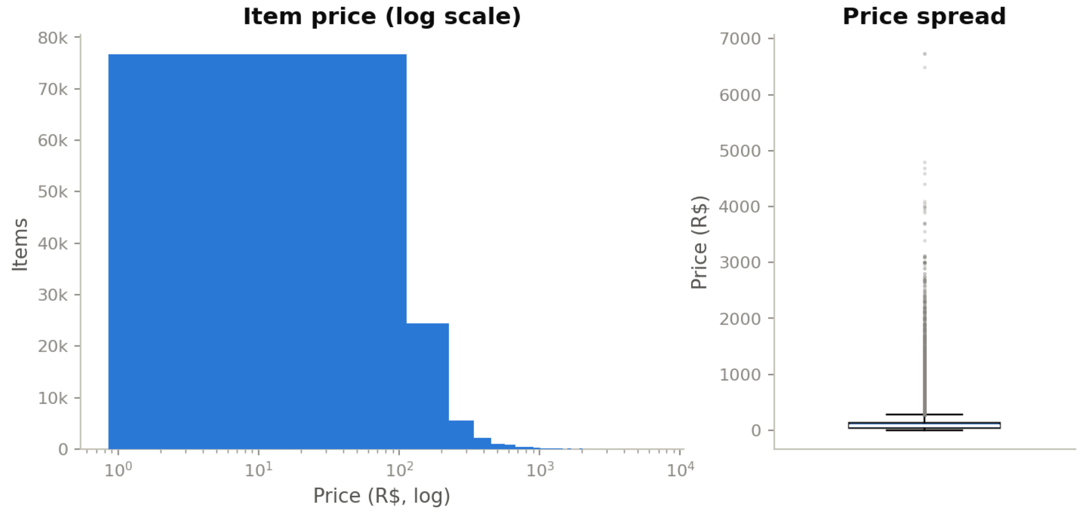

Figure 6. Item price on a log scale (left) and its spread (right). A genuine premium tail, not data errors.

5.6. Delivery Performance

Median delivery time (purchase to receipt) is 10 days, with a mean of 12.6 days pulled up by a long right tail. Measured against the marketplace's own estimated delivery date, **92% of orders arrive on time or early and 8% arrive late**, the operational SLA metric at the heart of Section 6.

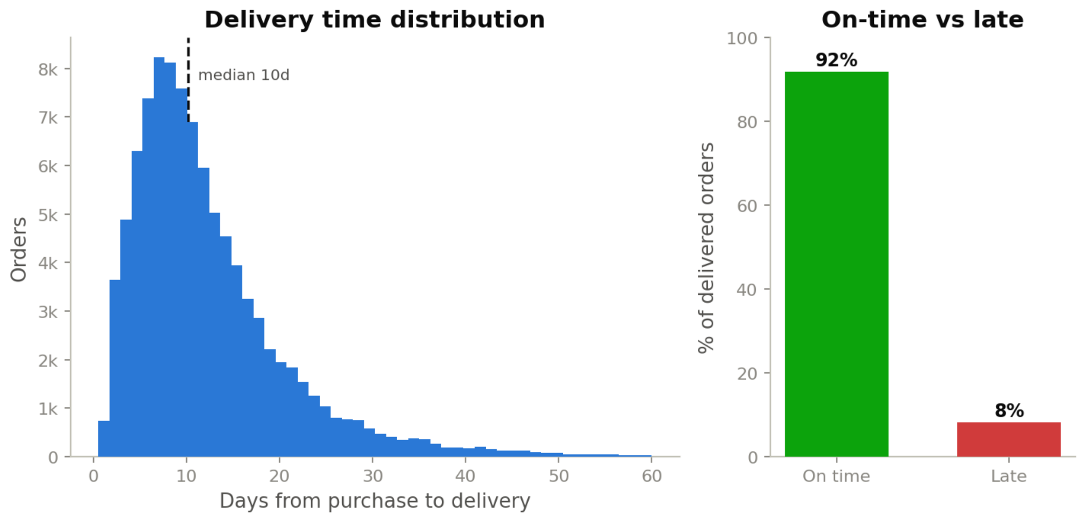

Figure 7. Delivery-time distribution (left) and on-time vs. late share (right).

5.7. Product Categories & Geography

Bed/bath/table, health & beauty and sports & leisure are the top categories by item volume, together roughly a quarter of all items. Geographically the marketplace is heavily concentrated: São Paulo state alone accounts for ~42% of orders, more than the next three states combined, an important weighting factor for any state-level comparison.

5.8. Relationships Between Variables

A correlation check across order-item and product-dimension fields shows `freight_value` correlates moderately with product weight ($r \approx 0.61$), heavier items cost more to ship, while price and freight are also positively related ($r \approx 0.41$). Photo count shows negligible correlation with price or freight, so richer listings do not, on their own, command higher prices.

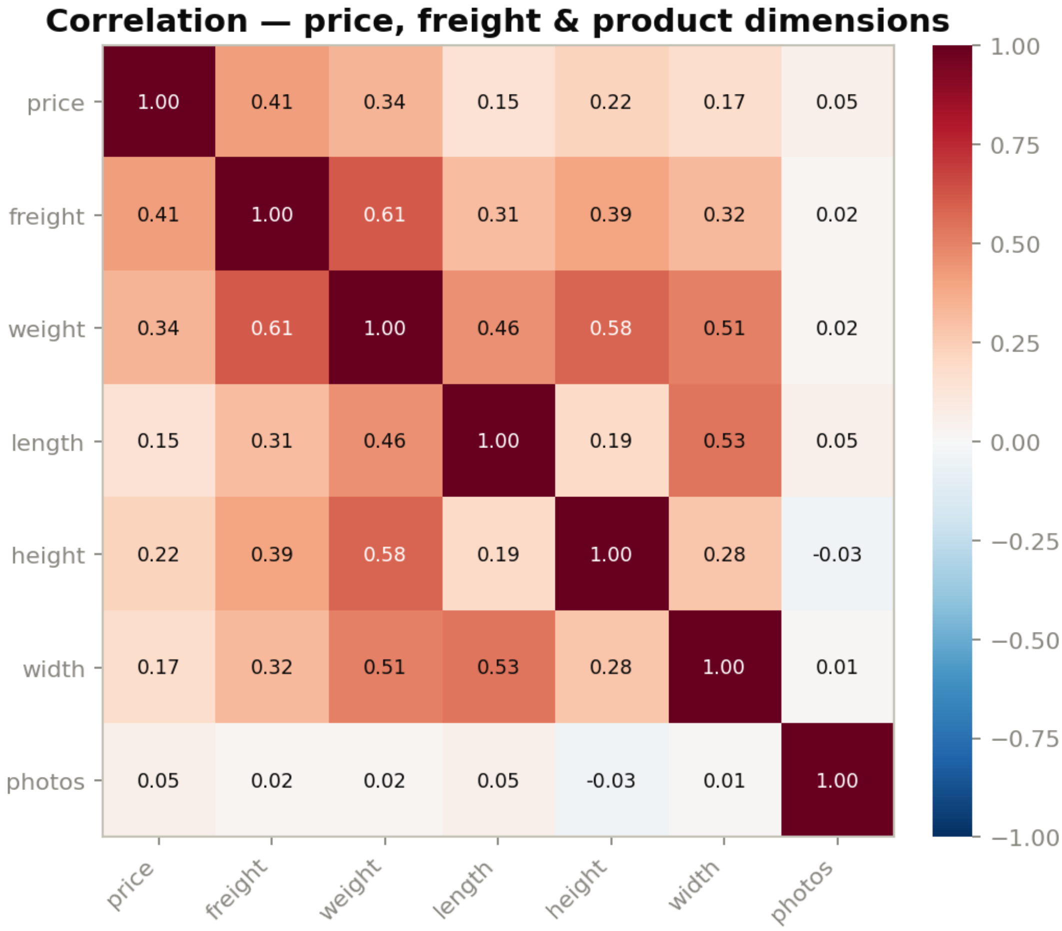

Figure 8. Correlation matrix, price, freight and product dimensions.

6. Explanatory Analysis - Answering the Questions

6.1. Q1 - Delivery Performance is the Dominant Driver of Satisfaction

Splitting delivered orders into on-time and late groups produces the sharpest result in the entire dataset. The contrast is not marginal, it is the difference between a promoter and a detractor.

Delivery status	Orders	Avg rating	Neg.-review rate	One-star rate
On time	88,644	4.29★	9.2%	6.6%
Late	7,825	2.57★	54.0%	46.2%

Table 7. On-time vs. late delivery outcomes. Lateness multiplies negative reviews five-to-six fold.

A late delivery does not shave a fraction off satisfaction. It **roughly halves** the average rating (4.29 → 2.57) and multiplies the one-star rate **seven-fold** (6.6% → 46.2%). This is the single most actionable signal in the data.

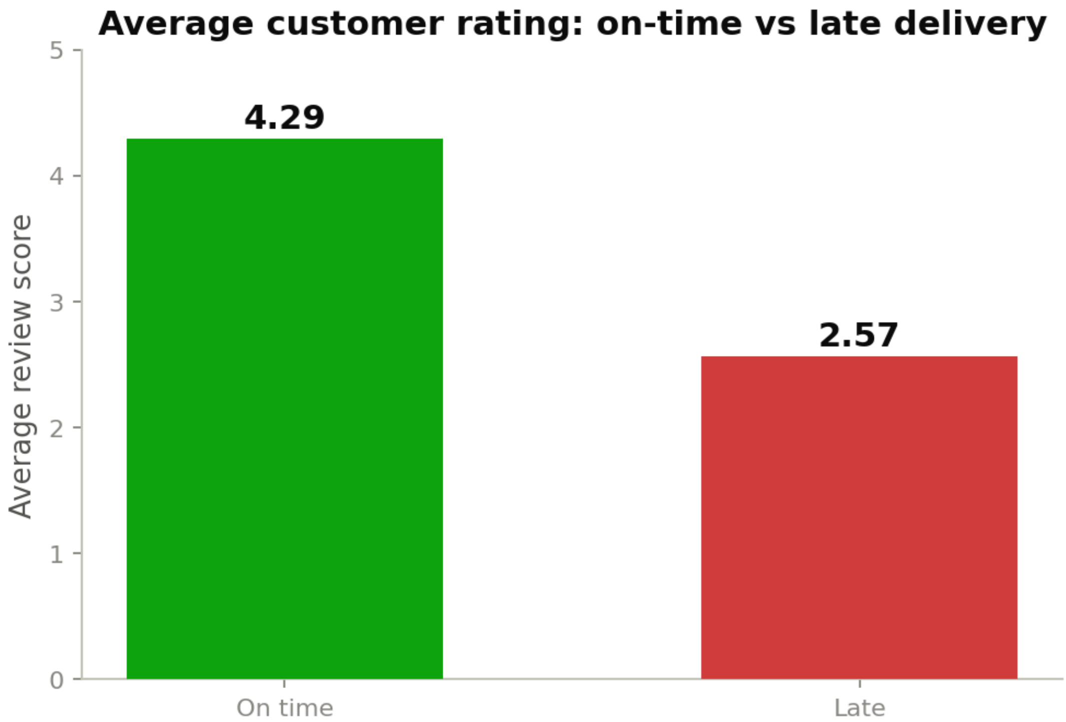

Figure 9. Average customer rating: on-time vs. late delivery. A late order roughly halves the score.

The relationship is continuous. The delivery satisfaction link also holds across the full distribution of delivery timing, not just the on-time/late binary. Because the marketplace pads its delivery estimates, orders of every rating tend to arrive ahead of the promised date on average but the size of that safety margin shrinks steadily as satisfaction falls.

- **Five-star orders** beat the estimated delivery date by nearly two weeks on average.
- **One-star orders** arrive only about three days ahead of the estimate and frequently spill past it altogether.
- This means the estimated delivery date itself functions as a *satisfaction threshold*: customers feel the miss acutely even when the absolute delivery time is reasonable.

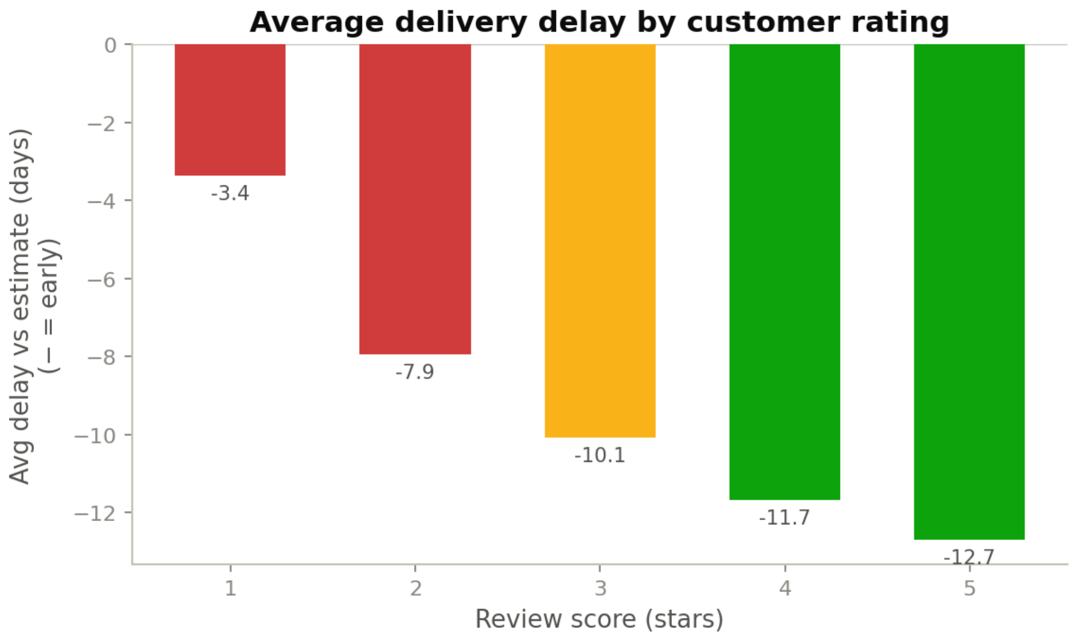

Figure 10. Average delivery delay versus estimate, by review score (negative = early).

Implication. On-time delivery is the **primary customer-experience KPI** for leadership to manage as the marketplace grows. Tightening estimated-delivery accuracy and prioritising the delivery promise ahead of catalogue expansion is the highest-ROI intervention available. Logistics capacity and carrier partnerships, particularly in underserved regions, are the investment levers.

6.2. Q2 - Where to Invest and Where to Intervene, by Category

To turn category performance into a decision, each category with ≥100 orders is placed on a **2×2 matrix** of revenue against average satisfaction, split at the median of each. This yields four action groups:

Quadrant	Meaning	Action	Count
Invest	High revenue, high satisfaction	Grow - these are the healthy engines	15
Intervene	High revenue, low satisfaction	Fix - big earners hurting the brand	11
Maintain	Low revenue, high satisfaction	Nurture - happy niches to grow	11
Review	Low revenue, low satisfaction	Re-evaluate - limited upside	15

Table 8. The category strategy framework used in the bubble-chart figure below.

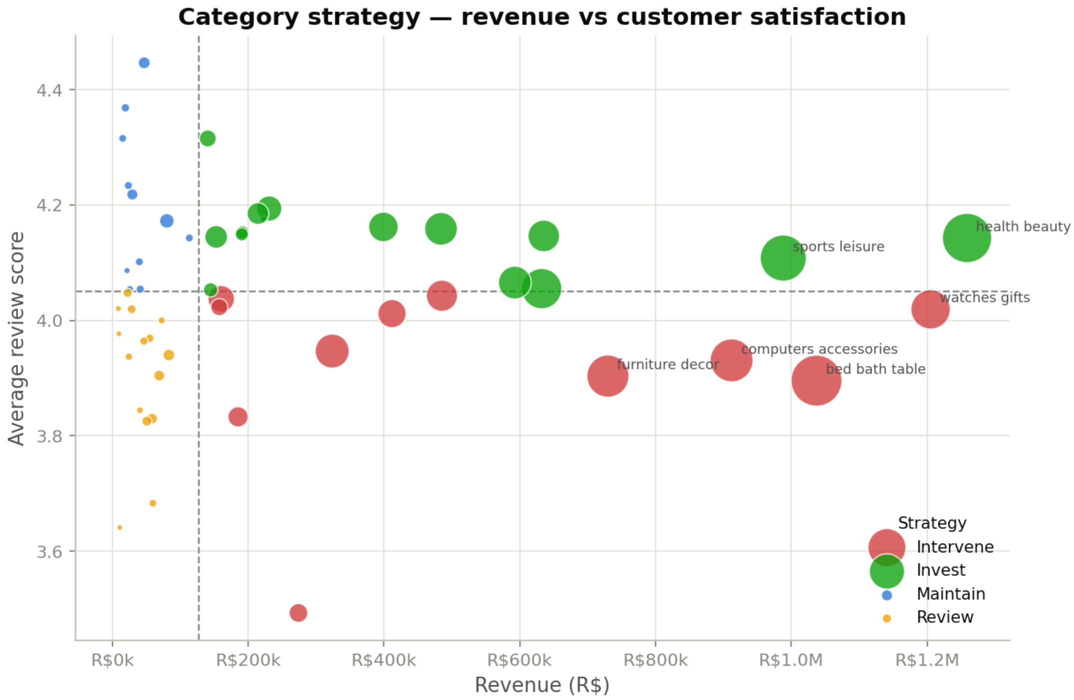

Figure 11. Category strategy, revenue (x) vs. satisfaction (y); bubble size = order volume.

Priority: the Intervene quadrant. These 11 categories already earn significant revenue, so every point of satisfaction recovered is high-value. They include several of the largest categories on the platform:

Category	Revenue (BRL)	Avg rating	One-star rate
watches_gifts	1,205,006	4.02⋆	12.6%
bed_bath_table	1,036,989	3.90⋆	14.5%
computers_accessories	911,954	3.93⋆	15.0%
furniture_decor	729,762	3.90⋆	15.1%
garden_tools	485,256	4.04⋆	12.8%
office_furniture	273,961	3.49⋆	21.0%

Table 9. Highest-revenue intervene categories. *office_furniture* is highlighted as the priority intervention target.

> **office_furniture** is the single clearest intervention target: it combines meaningful revenue with the **worst average rating** (3.49⋆) and a **21% one-star rate** driven by freight complexity (avg BRL 40.4 vs BRL 16–19 for most categories) and likely assembly/quality issues.

For balance, the figure below lists the lowest-rated categories overall; office_furniture in particular combines meaningful revenue with the worst satisfaction, making it the single clearest intervention target.

6.3. Q3 - The Regional Risk Map

Delivery risk is strongly regional. São Paulo, 42% of orders, is both fast and well-rated. The pain is concentrated in the **North and Northeast**, where the highest late rates coincide with the lowest ratings and the longest delivery times. These are logistics problems, not demand problems, and they are exactly where an on-time-delivery programme would move the satisfaction needle most.

State	Orders	Late rate	Avg rating	Avg delivery (days)
AL - Alagoas	397	23.9%	3.84★	24.5
MA - Maranhão	717	19.6%	3.84★	21.5
PI - Piauí	476	15.9%	4.00★	19.5
CE - Ceará	1,279	15.4%	3.94★	21.3
SP - São Paulo (reference)	40,493	5.9%	4.25★	8.8

Table 10. Worst-performing delivery states vs. São Paulo for reference.

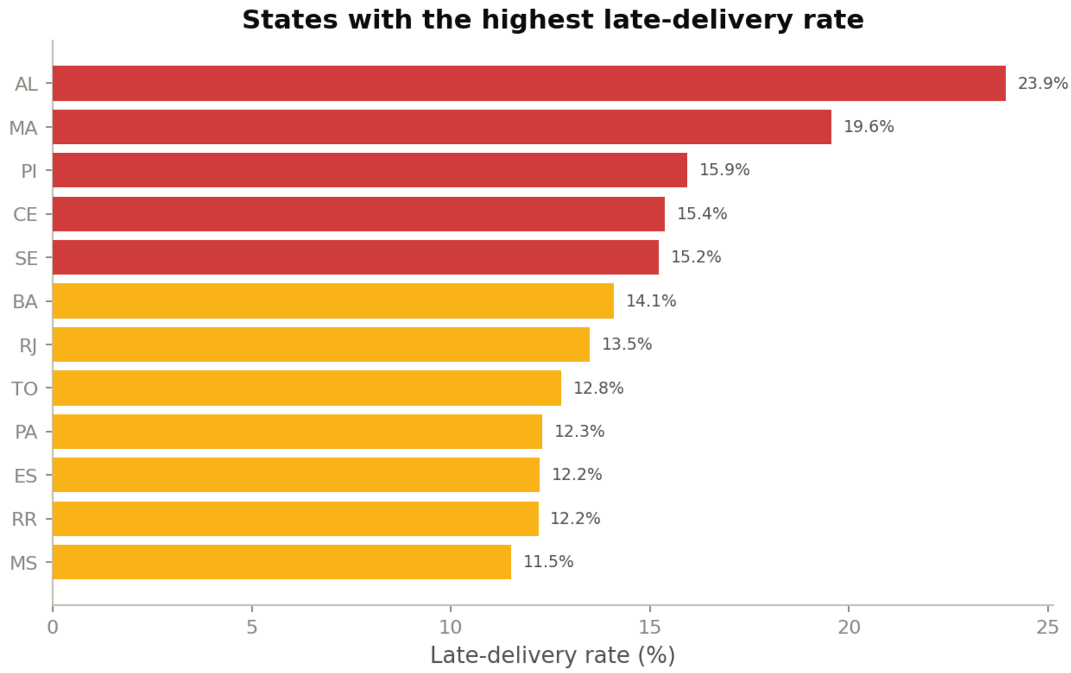

Figure 12. States with the highest late-delivery rate.

Key observations.

- The North/Northeast states face delivery times **2–3× longer** than São Paulo, driven by distance from the seller concentration in the South/Southeast and limited carrier infrastructure.
- Revenue is as concentrated as risk is dispersed: the top 12 states by revenue are almost entirely in the South/Southeast, meaning the regions with the worst delivery experience are also the least commercially prioritised.
- Setting **region-specific delivery estimates** so promises stay credible would partially decouple satisfaction from logistics performance in the short term, while investment

in carrier partnerships builds the longer-term fix.

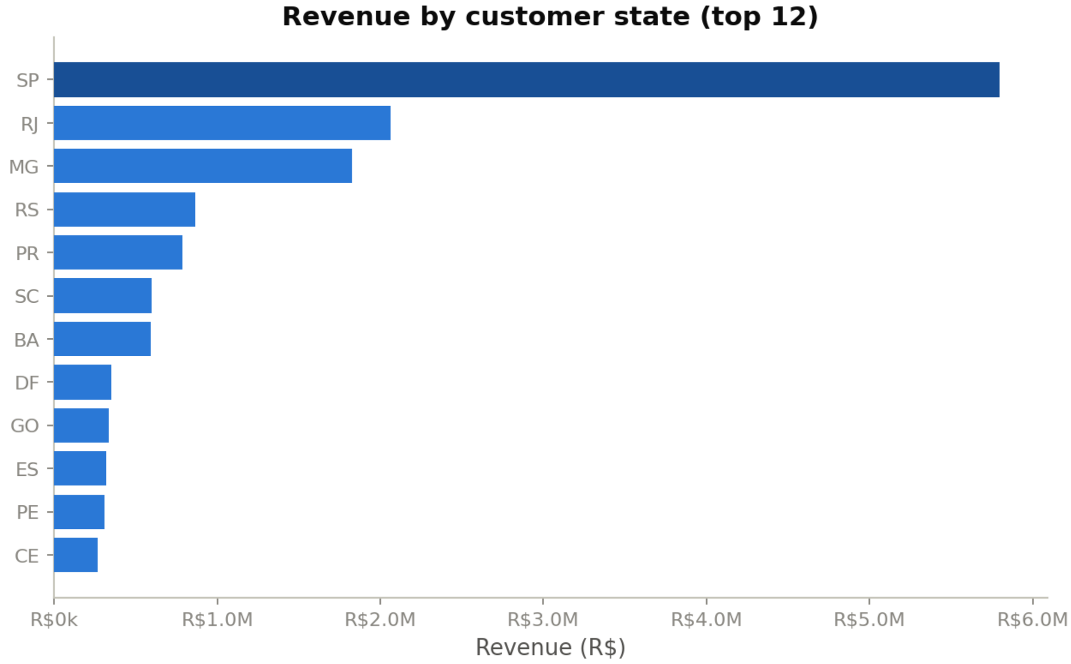

Figure 13. Revenue by state (top 12). Revenue is as concentrated as risk is dispersed.

> **Seller geography amplifies the problem:** approximately 60% of sellers are located in São Paulo state. Diversifying the seller base geographically or investing in regional fulfilment partnerships would reduce transit distances and cut late-delivery rates in the North/Northeast without waiting for national carrier infrastructure to improve.

6.4. Q4 - Seller Risk is Concentrated, Not Systemic

A crackdown on 'slow sellers' is only justified if poor performance is widespread and it is not. Among the **429 sellers with 50 or more orders**, the median rating is 4.1 and the median late rate is 6.8%, in line with the marketplace as a whole. Applying a strict risk screen (100+ orders, rating below 3.5 and late rate above 20%) flags **just one seller**.

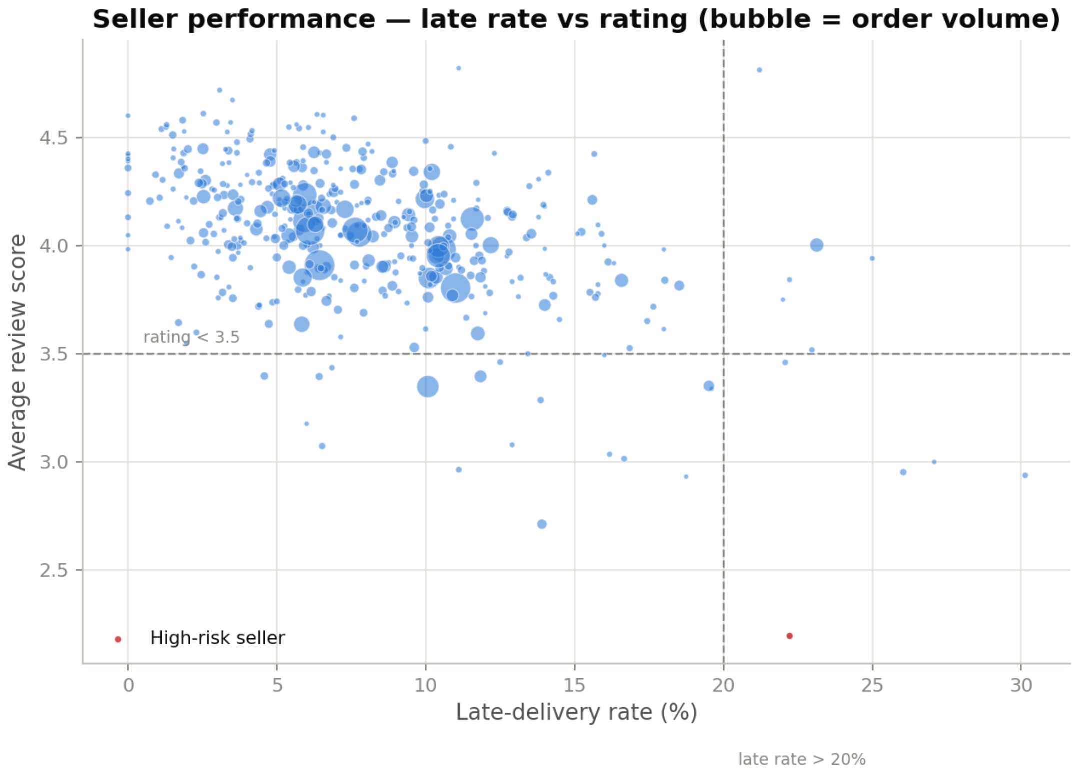

Figure 14. Seller late rate vs. rating (bubble = order volume). The high-risk quadrant is nearly empty.

> **Implication:** the catalogue does not need to be pruned broadly. A small number of underperformers can be coached or delisted with negligible impact on product selection protecting both quality and growth.

6.5. Q5 - Freight Cost and Satisfaction

Freight cost shows a **mild negative association** with rating: lower-rated orders carry somewhat higher average freight. This is largely a by-product of geography and delivery, the distant regions that incur higher freight costs are the same ones suffering late deliveries, rather than customers directly penalising shipping charges. Freight is therefore best treated as a lever on the delivery problem rather than a separate satisfaction driver.

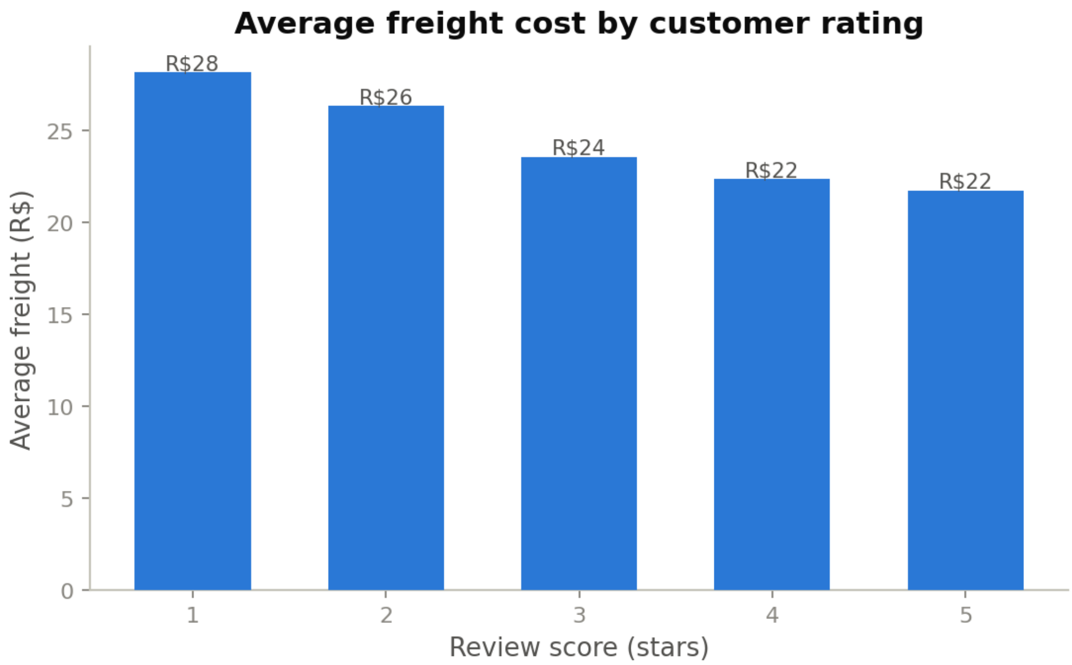

Figure 15. Average freight cost by customer rating.

6.6. Q6 - Efficient Channels for Seller Acquisition

On the acquisition side, channel volume and channel efficiency diverge sharply. Organic search is the largest lead source by volume, but conversion rates tell a different story:

Channel	Lead volume	Conv. rate	Recommendation
Unknown / direct	High	16.7%	Investigate and attribute - likely high-intent traffic
Paid search	Medium	12.3%	Scale - highest attributable ROI
Organic search	High	11.8%	Maintain - solid and cost-efficient
Social	Medium	5.6%	Fix the funnel before scaling spend

Table 11. Marketing channel volume vs. conversion efficiency. Spend should follow conversion rate, not raw lead count.

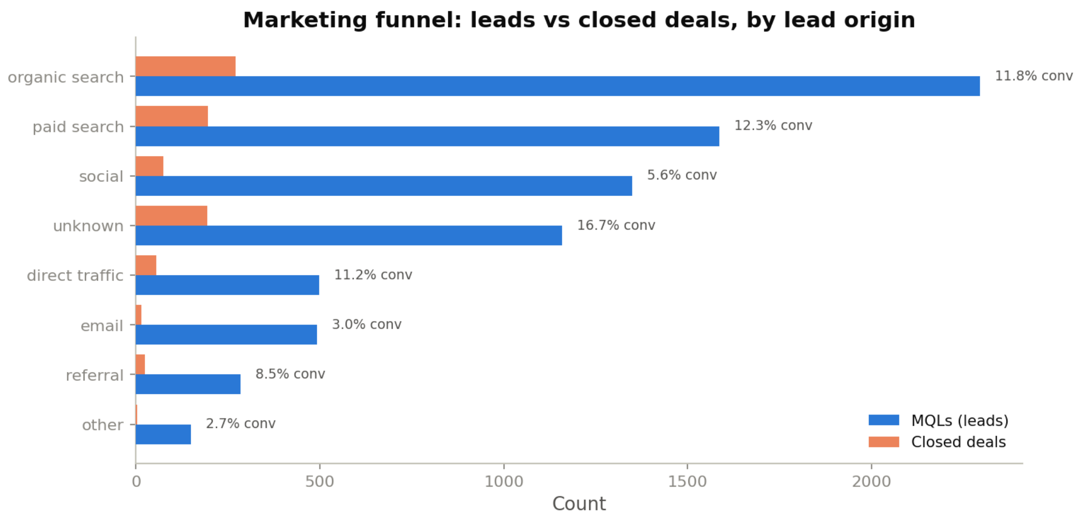

Figure 16. Marketing funnel, MQL volume vs. closed deals by lead origin, with conversion rate.

7. Interactive Marketplace Dashboard

The analytical findings are delivered as an interactive Streamlit dashboard so that leadership can explore satisfaction and risk themselves rather than reading static charts. The dashboard reads the same analytical tables produced by the pipeline, so it is always consistent with this report.

7.1. Structure

- **Global KPI header**, orders, revenue, customers, average rating, late-delivery rate, negative-review rate and average order value, all responding to the filters.
- **State filter** in the sidebar lets a viewer narrow the entire dashboard to one or more states to see why a region stands out.
- **Overview** tab, monthly sales trend, top categories and revenue by state.
- **Category Strategy** tab, the Invest/Intervene/Maintain/Review scatter and a sortable strategy table.
- **Delivery & Satisfaction** tab, on-time vs. late ratings, delay by score, and delivery-time distributions.
- **Regional** tab, revenue and a late-rate-vs-rating scatter, with best/worst callouts.
- **Seller Performance** tab, the risk scatter and a high-risk seller table.

7.2. Running the Dashboard

From the repository root: install dependencies with pip, then launch with Streamlit. The app opens in the browser and is fully interactive.

```
pip install -r requirements.txt
streamlit run app.py
```

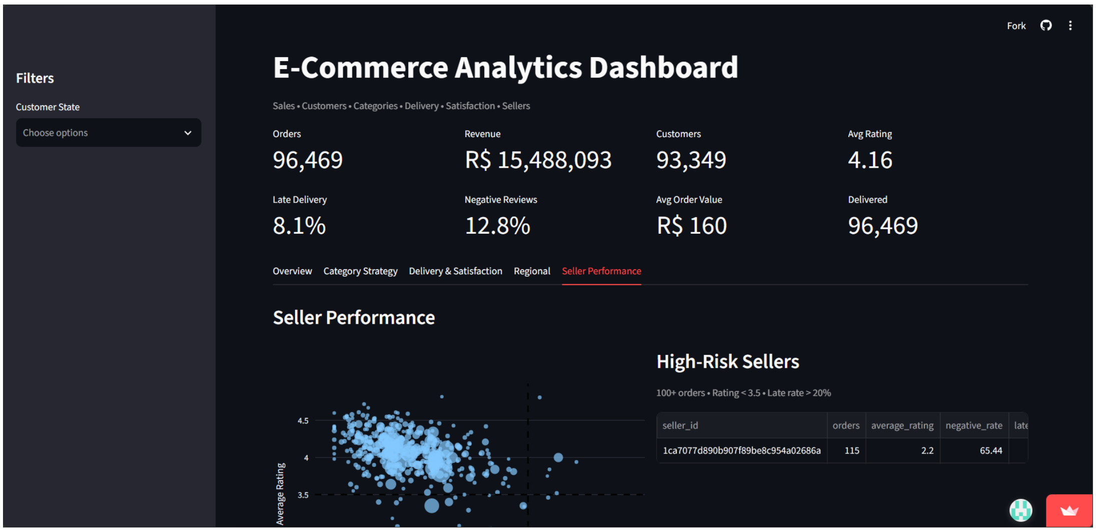

Figure 17. Dashboard Overview

8. Key Insights & Business Implications

Bringing the six questions together, the analysis converts the growth-vs-satisfaction dilemma into a small set of concrete, evidence-backed actions.

8.1. From Insight to Action

Insight	**Recommended action**
Late delivery collapses satisfaction (4.29 → 2.57 ⋆; negatives 9% → 54%)	Manage on-time delivery as the primary customer-experience KPI. Tighten estimated-delivery accuracy and prioritise the delivery promise ahead of catalogue expansion.
11 high-revenue categories rate below the marketplace median	Run category-level quality and delivery interventions on watches_gifts, bed_bath_table, computers_accessories and furniture_decor first; highest revenue at risk.
Delivery risk is regional, concentrated in the North/Northeast	Invest in logistics capacity and carrier partnerships for AL, MA, PI and CE; set region-specific delivery estimates to keep promises credible.
Seller risk is concentrated in a handful of sellers, not systemic	Coach or delist the few genuine underperformers instead of a broad crackdown protecting catalogue breadth and growth momentum.

Insight	Recommended action
Acquisition channels differ sharply in efficiency (search ∼12% vs. social 5.6%)	Reallocate seller-acquisition spend toward search; diagnose and fix the social-channel funnel before scaling it.
Growth has plateaued at ∼6–7k orders/month with clear seasonal peaks	Plan logistics capacity around Black Friday and similar peaks when late-delivery risk and its satisfaction cost are highest.

Table 13. From insight to action; recommendations for leadership and operations.

Together these actions resolve the leadership dilemma: the marketplace can keep growing its catalogue because seller quality is not a systemic problem, provided it invests in the **delivery experience**, especially in the underserved regions and highest-revenue categories, which is where satisfaction is actually won or lost.

9. Constraints, Limitations & Future Improvements

9.1. Limitations

- The data ends in 2018, so findings describe historical dynamics rather than current absolute numbers.
- Satisfaction analysis excludes cancelled and undelivered orders, which likely carry their own negative sentiment.
- The delivery satisfaction link is very strong and directionally clear, but the analysis is **observational**; a controlled test would be needed to quantify causal lift.
- With São Paulo accounting for 42% of orders, national averages are dominated by one state; state-level breakdowns are reported throughout for this reason.

9.2. Future Improvements

- Text mining of review comments to explain *why* late or low-rated orders disappoint, beyond the score alone.
- A **predictive late-delivery model** to flag at-risk orders at checkout and set honest delivery estimates dynamically.
- Customer-lifetime and repeat-purchase analysis to quantify the revenue cost of a one-star experience over time.
- Choropleth mapping using the geolocation table for a richer regional view in the interactive dashboard.

10. Conclusion

> The marketplace's growth-versus-satisfaction dilemma has a clearer resolution than gut feel suggests. Satisfaction is overwhelmingly driven by delivery timing: an on-time order is a 4.3-star experience; a late one is a 2.6-star experience. Because poor sellers are rare and the catalogue is broadly healthy, the company does not have to choose between growth and quality. It has to **protect and extend the delivery promise**, focus quality interventions on the eleven high-revenue categories that under-deliver, and close the regional delivery gap in the North and Northeast. Do that, and the marketplace can keep onboarding sellers and growing sales without breaking the customer experience.

11. Technical Appendix

11.1. Repository Structure

The complete code repository, all scripts, analytical outputs, figures and the dashboard needed to reproduce this work, is available at `github.com/surya101p/DVD_23f2004625`. The final presentation is available `here`.

Path	Contents
`DVDProject_analysis.py`	End-to-end cleaning + analysis pipeline; writes the 8 analytical tables
`generate_figures.py`	Regenerates all 20 report/presentation figures from data
`app.py`	Streamlit interactive dashboard
`analysis_output/*.csv`	Analytical tables (order, category, seller, regional, delivery, strategy)
`figures/*.png`	All figures used in this report
`requirements.txt`	Python dependencies
`Technical_Report.pdf`	This report

Table 14. Repository layout for reproducing the work.

11.2. How to Reproduce

1. Place the raw dataset folders (E-Commerce Dataset, Marketing Funnel) alongside the scripts.
2. Run the analysis pipeline to produce the analytical tables in `analysis_output/`.
3. Run `generate_figures.py` to rebuild every figure in `figures/`.
4. Run the Streamlit app to explore the dashboard.

11.3. Analytical Output Tables

- **order_analysis.csv**, order-level dataset (delivery metrics + review + value + customer).
- **category_performance.csv / category_strategy.csv**, category revenue, satisfaction and quadrant.

- **seller_performance.csv**, per-seller rating, negative rate and delivery metrics.
- **regional_performance.csv**, per-state revenue, rating, late rate and delivery days.
- **delivery_satisfaction.csv**, the on-time vs. late summary behind the delivery-rating figure in Section 6.1.
- **delivered_orders.csv / item_analysis.csv**, cleaned intermediate datasets.

11.4. Environment

Python 3, pandas, NumPy, matplotlib, seaborn (analysis and figures); Streamlit and Plotly (dashboard). Exact versions are pinned in `requirements.txt`.

12. Team Contributions & Work Log

Group 10's members and their contributions are documented below for transparent and fair assessment. *Edit the responsibilities and contributions to match what each member actually did.*

Member (Roll no.)	Primary responsibilities	Key contributions
Astitva Agarwal (23f2004131)	Exploratory analysis & insights	Conducted initial EDA, identified key trends and supported interpretation of findings.
Nishant Kumar (22f3003042)	Presentation & visualization	Finalised the PPT, selected key visuals and structured the final presentation narrative.
Yash Kumar (22f3000472)	Data preparation & analysis	Cleaned and prepared the data, handled inconsistencies and supported the analysis pipeline.
Diksha Singhal (21f1002910)	Technical report & documentation	Finalised the technical report, documenting methodology, findings, implications and limitations.
Surya Prakash (23f2004625)	Dashboard development	Developed and finalised the interactive dashboard and integrated the key project findings.

Table 15. Individual contributions.

12.1. Weekly Work Log

Week	Planned goal	Status / notes
Week 1	Understand data; restate goal; early EDA; concept note	Completed. Dataset and objectives were reviewed, initial EDA was conducted and the concept note was prepared.
Week 2	Cleaning; detailed EDA; explanatory charts; mid-week summary	Completed. Data was cleaned and analysed, key findings were identified and explanatory visualizations were prepared.
Week 3	Dashboard; technical report; final presentation; review	Completed. Dashboard, technical report and PPT were finalised, followed by team review and refinement of all deliverables.

Table 16. Weekly progress against the project roadmap.